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Cambridge, Massachusetts

Divert-feasible lunar landing under navigational uncertainty

Yana Lishkova[1], Abraham Vinod[2], Stefano Di Cairano[2], and Avishai Weiss[2] [12]

[1]Department of Engineering Science, University of Oxford, UK; intern at MERL during the development of this work.
[2]Mitsubishi Electric Research Laboratories (MERL), MA, 02139, USA

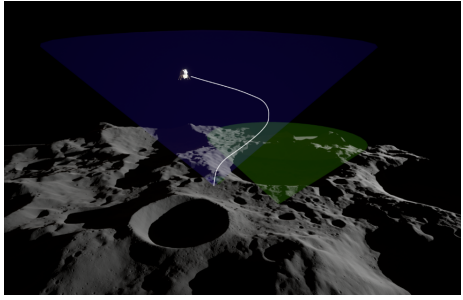
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Motivation

Powered descent guidance (PDG)





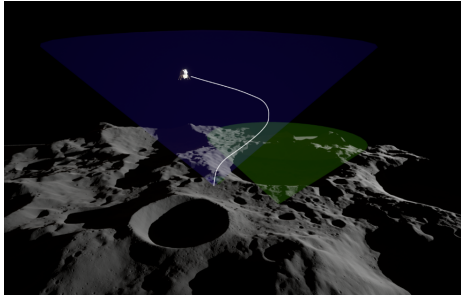
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Powered descent guidance (PDG) challenges

- complex missions
- high degree of autonomy
- unexpected hazards





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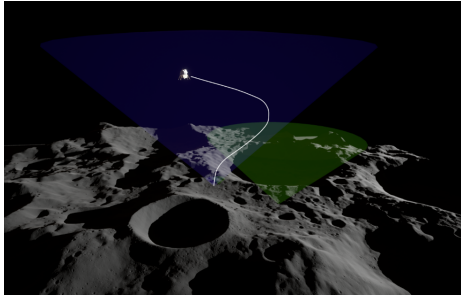
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Computational challenges:

- real-time trajectory computation
- nonconvex model and state/control constraints





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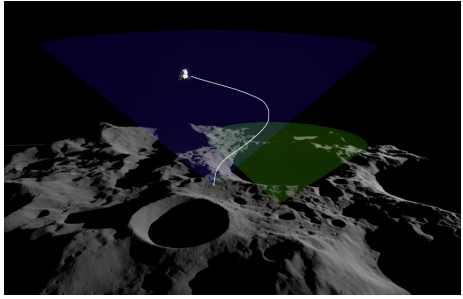
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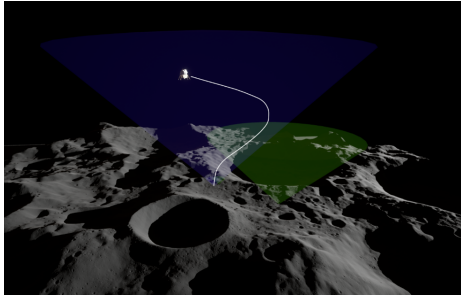
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Computational challenges:

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- nonconvex model and state/control constraints
- presence of navigation noise





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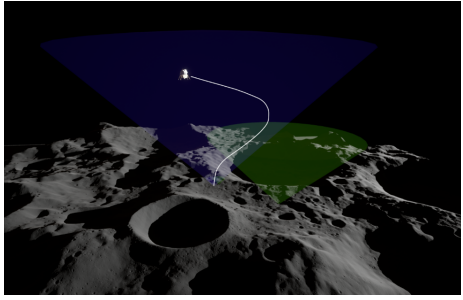
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Motivation

Problem 1 (Nominal trajectory design):
Design a tractable optimization formulation for constraint-admissible PDG under navigational uncertainty with greater than or equal to user-specified safety probability.





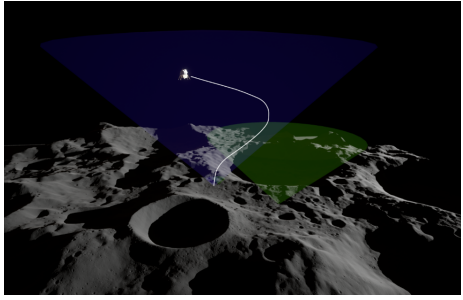
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Motivation

Problem 1 (Nominal trajectory design):
Design a tractable optimization formulation for constraint-admissible PDG under navigational uncertainty with greater than or equal to user-specified safety probability.

Problem 2 (Hazard avoidance):
If a hazard is detected, identify a reachable alternative landing site and use a modified version optimization-based problem to divert to the selected site

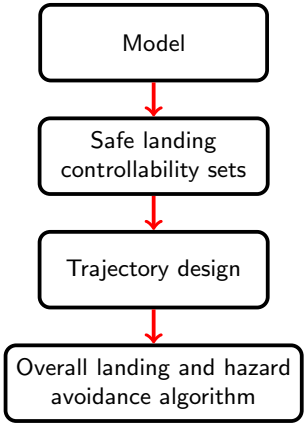


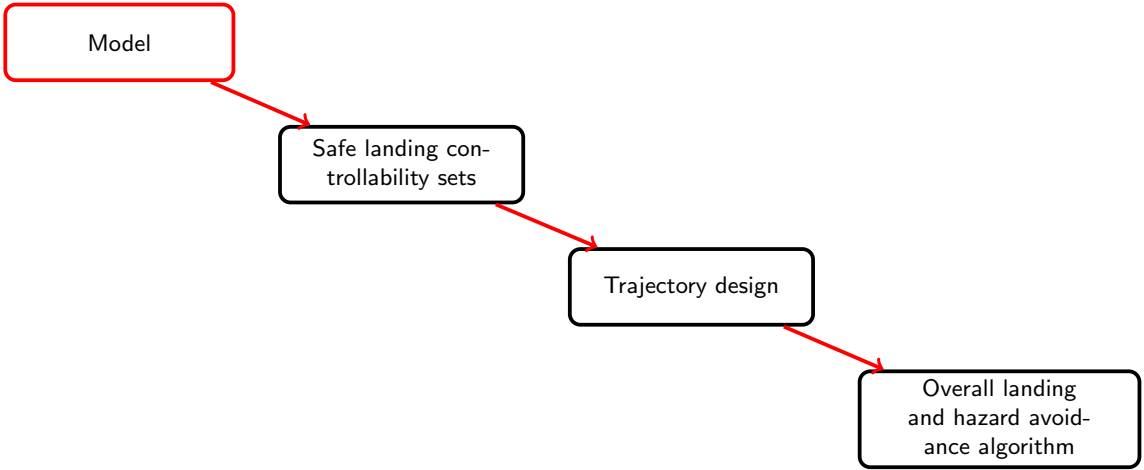


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Talk overview

- Motivation
- System model and constraints
  - Uncertainty modeling
  - Formulation
  - Convexification and discretization
- Safe stochastic controllability set computation
- Landing trajectory design
- Overall divert-feasible PDG algorithm under navigational uncertainty




Model
Safe landing controllability sets
Trajectory design
Overall landing and hazard avoidance algorithm

System Modelling

$$\ddot{r}(t) = \frac{T_{\mathrm{c}}(t)}{m(t)} + g$$

$$\dot{m}(t) = -\beta \|T_{\mathrm{c}}(t)\|$$

$r \in \mathbb{R}^3$ - the translational position of the lander
$m \in \mathbb{R}_+$ - the mass of the lander
$T_{\mathrm{c}} \in \mathbb{R}^3$ - the net thrust applied to the lander

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State space model

$$x(t) \triangleq \begin{bmatrix} r(t) \\ \dot{r}(t) \end{bmatrix}, \quad u(t) \triangleq \frac{T_c(t)}{m(t)}$$

$=>$ the translational dynamics become

$$\dot{x}(t) = Ax(t) + Bu(t) + G$$

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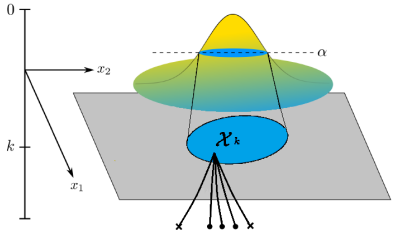
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Uncertainty Modelling

Define a noisy state

$$y(t) = x(t) + \nu(t), \quad \nu(t) \sim \mathcal{N}(0, \Sigma_\nu)$$



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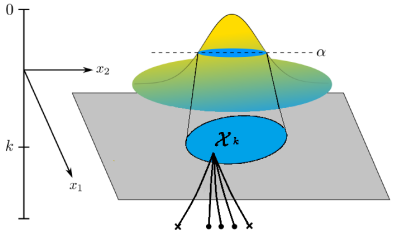
Initial-time Gaussian noise $\nu(0)$ reformulated as [4]

$$\mathcal{E}_\alpha : \{\nu(0) \mid \nu(0)^\intercal \Sigma^{-1} \nu(0) \leq F^{-1}_{\chi^2(6)}(\alpha)\},$$

$$\text{i.e } \mathbb{P}\{\nu(0) \in \mathcal{E}_\alpha\} = \alpha$$

where $\chi^2(6)$ is a chi-squared random variable with six degrees of freedom, and $F_{\chi^2(6)}(\cdot)$ is its cumulative distribution function.

[4] J. D. Gleason, A. P. Vinod, and M. M. Oishi, "Lagrangian approximations for stochastic reachability of a target tube," Automatica, vol. 128, p. 109546, 2021



Uncertainty Modelling

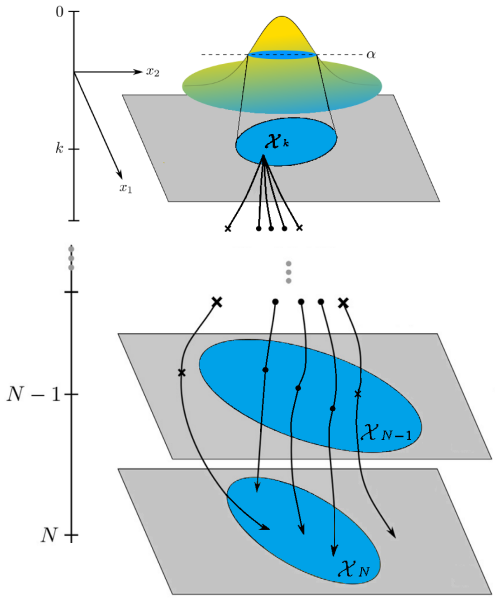
$$y(t) = x(t) + \nu(t), \;\; \nu(t) \sim \mathcal{N}(0, \Sigma_\nu)$$

$$\mathcal{E}_\alpha : \{\nu(0) \mid \nu(0)^\intercal \Sigma^{-1} \nu(0) \leq F^{-1}_{\chi^2(6)}(\alpha)\}$$

Thus, given an estimate $y(0)$:

$$\mathcal{X}(0) = \{y(0)\} \oplus \mathcal{E}_\alpha$$

contains all possible true initial states $x(0)$ with probability α



Uncertainty Modelling

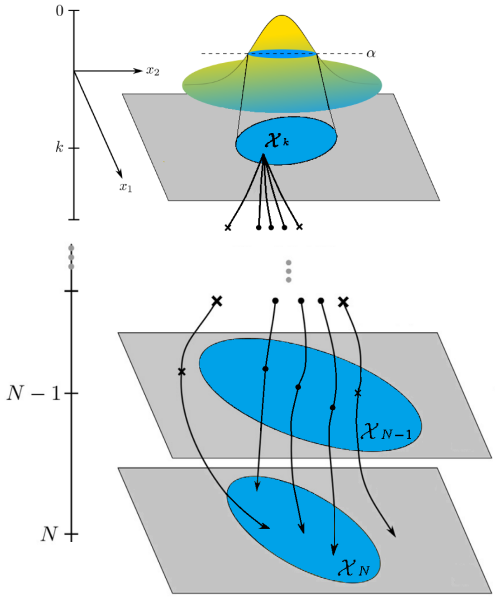
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The evolution of the possible true trajectories $x(t)$ evolving from $\mathcal{X}(0)$ under a given open loop control sequence is given by

$$x(t) \in \mathcal{X}(t) : \{x \mid (\bar{x}(t) - x)^\intercal S(t) (\bar{x}(t) - x) \leq 1\},$$



Uncertainty Modelling

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$$S(t) = \Phi(0,t) S(0) \Phi(0,t)^\intercal, \quad S(0) = \frac{1}{R^2} \Sigma^{-1},$$

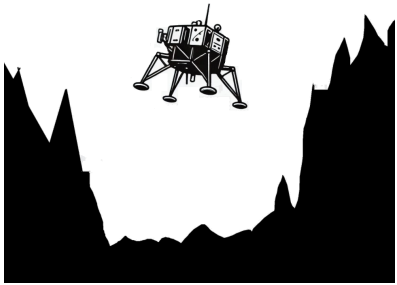
$$\dot{\bar{x}}(t) = A\bar{x}(t) + Bu(t) + G, \quad \bar{x}(0) = y(0),$$

where $R^2 = F^{-1}_{\chi^2(6)}(\alpha)$, and $\Phi(0,t)$ is the state transition matrix for the state space model.

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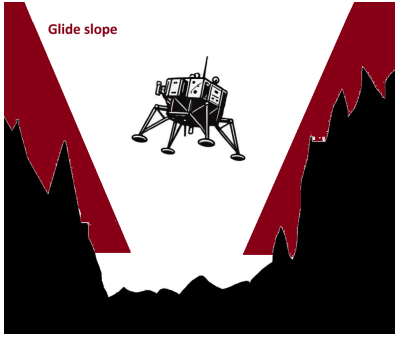
Constraints

- mass range: $m_{\mathrm{d}} \leq m(t) \leq m_{\mathrm{w}}$
- maximum velocity constraint: $\|\dot{r}(t)\| \leq v_{\max}$



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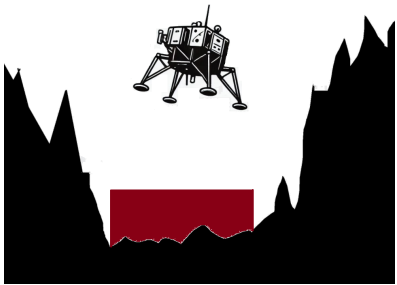
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- gliding constraint (nonconvex): $\|r(t) - r_{\mathrm{ls}}\| \cos(\gamma) \leq \hat{k}_{\mathrm{s}}^{\mathsf{T}} r(t)$



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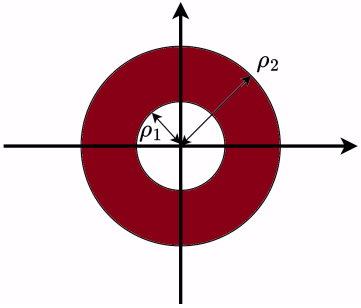
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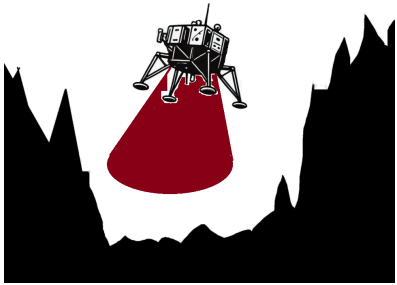
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$$H(\bar{x}(t) - x_{\mathrm{ls}}) \leq f$$

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$$H(\bar{x}(t) - x_{\mathrm{ls}}) \leq f \quad \rightarrow \quad \|S(t)^{1/2}[H]_i\| \leq -[H]_i(\bar{x}(t) - x_{\mathrm{ls}}) + [f]_i \;\; i = 1, ..., n_H$$

where $x(t) \in \mathcal{X}(t) : \{x \mid (\bar{x}(t) - x)^{\intercal} S(t)(\bar{x}(t) - x) \leq 1\}$

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Final formulation

Final formulation

Final model

$$S(t) = \Phi(0,t)S(0)\Phi(0,t)^\intercal, \quad S(0) = \frac{1}{R^2}\Sigma^{-1},$$
$$\dot{\bar{x}}(t) = A\bar{x}(t) + Bu(t) + G, \quad \bar{x}(0) = y(0),$$
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Constraints

- mass range
- tightened state constraints
  (maximum velocity, glideslope, landing region)
- torque bounds (nonconvex)
- pointing constraint (nonconvex)

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In an optimal control problem these can be

- **convexified** using the standard lossless convexification approach [3,5-7]

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[7] B. Açıkmeşe et al., "Enhance- ments on the convex programming based powered descent guidance algorithm for Mars landing," in AIAA/AAS astrodynamics specialist conference and exhibit, 2008, p. 6426.

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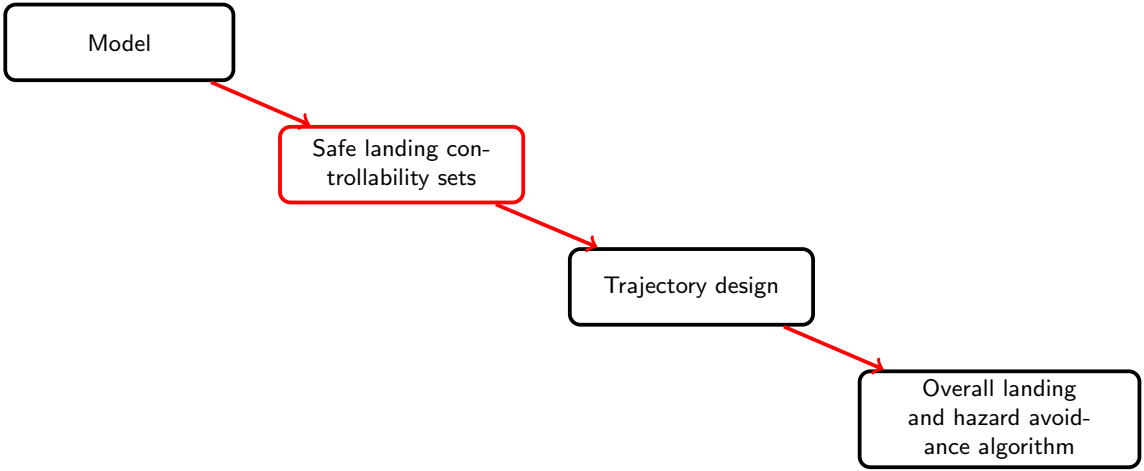
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  $\Rightarrow$ **optimality and feasibility guarantees**

- **discretized**


Model
Safe landing controllability sets
Trajectory design
Overall landing and hazard avoidance algorithm

Safe stochastic controllability sets (SSCS)

- **6DOF convex** sets of noisy state estimates from which the lander can be driven to a specified landing site with user-specified probability
- parametrized by the mass $m_{\mathrm{w}}^{\mathrm{s}} \in [m_{\mathrm{d}}, m_{\mathrm{w}}]$

Safe stochastic controllability sets (SSCS)

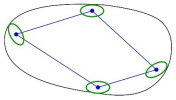
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- parametrized by the mass $m_{\mathrm{w}}^{\mathrm{s}} \in [m_{\mathrm{d}}, m_{\mathrm{w}}]$
- obtained in three stages [8,9]

$$\max_{\bar{x}(0), u, t_{\mathrm{f}}} \; [[\bar{x}(0)^{\intercal}, -\bar{x}(0)^{\intercal}]^{\intercal}]_i$$

s.t. for $\forall t \in [0, t_{\mathrm{f}}]$

the convexified dynamics

the convexified constraints



1) Initial set

[8] U. Eren, D. Dueri, and B. Açıkmeşe, "Constrained reachability and controllability sets for planetary precision landing via convex optimization," *Journal of Guidance, Control, and Dynamics*, vol. 38, no. 11, pp. 2067–2083, 2015
[9] N. Srinivas, A. P. Vinod, S. Di Cairano, and A. Weiss, "Lunar landing with feasible divert using controllable sets," in AIAA SCITECH 2024 Forum, 2024, p. 0324.

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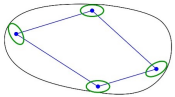
For $i = 1, ..., n_{\mathrm{facets}}$

$$\max_{\bar{x}(0), u, \sigma, t_{\mathrm{f}}} \quad \hat{n}_{\mathrm{facet},i}^{\mathsf{T}} \bar{x}(0)$$

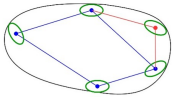
s.t. for $\forall t \in [0, t_{\mathrm{f}}]$

the convexified dynamics

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1) Initial set



2) Pushing along a face

[8] U. Eren, D. Dueri, and B. Açıkmeşe, "Constrained reachability and controllability sets for planetary precision landing via convex optimization," *Journal of Guidance, Control, and Dynamics*, vol. 38, no. 11, pp. 2067–2083, 2015
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Safe stochastic controllability sets (SSCS)

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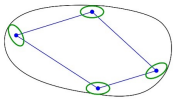
For $i = 1, ..., n_{\mathrm{facets}}$

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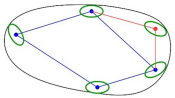
s.t. for $\forall t \in [0, t_{\mathrm{f}}]$

the convexified dynamics

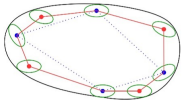
the convexified constraints



1) Initial set



2) Pushing along a face



3) Result

Safe stochastic controllability sets (SSCS)

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- parametrized by the mass $m_{\mathrm{w}}^{\mathrm{s}} \in [m_{\mathrm{d}}, m_{\mathrm{w}}]$
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- discretised case - pre-computed database of sets parametrized by (m_w^s, t_f^s)

Safe stochastic controllability sets (SSCS)

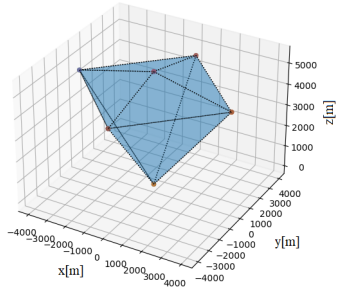
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- discretised case - pre-computed database of sets parametrized by (m_w^s, t_f^s)
- feasibility and safety guarantees

Safe stochastic controllability sets: Results

Moon & Lander parameters		Constraints specifications		Simulation parameters	
g	$[0, 0, -1.62]^\intercal m/s^2$	$x_{\text{ls}}^{\text{nominal}}$	$[0, 0, 0, 0, 0, 0]^\intercal$	Δt	10 s
m_{d}	900 kg	$v_{\max}$	300 m/s	α	0.7
ρ_1	1657.27 N	$\theta_{\max}$	25°	$3\sigma([\nu_0]_{[1:3]})$	$[3, 3, 3]^\intercal$ m
ρ_2	4419.39 N	$\hat{n}$	$[0, 0, 1]^\intercal$	$3\sigma([\nu_0]_{[4:6]})$	$[0.01, 0.01, 0.01]^\intercal$ m/s
I_{sp}	255 s	γ	75°	m_{fuel}	350 kg

Safe stochastic controllability sets: Results

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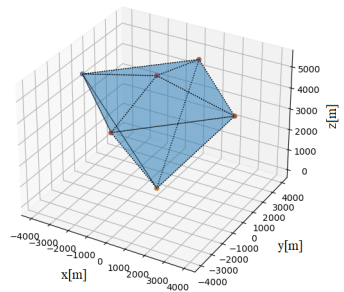


(a) $iter = 1$

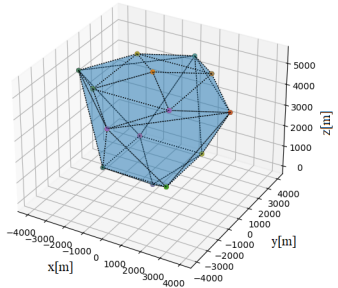
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Safe stochastic controllability sets: Results

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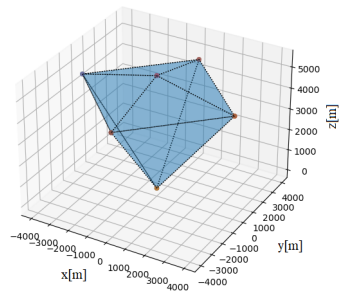


(b) $iter = 2$

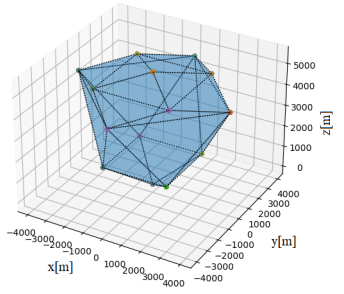
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Safe stochastic controllability sets: Results

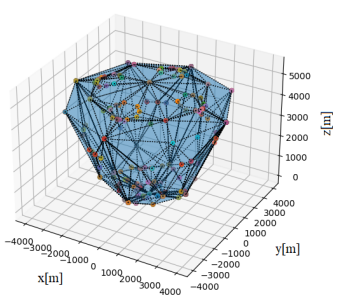
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I_{sp}	255 s	γ	75°	m_{fuel}	350 kg



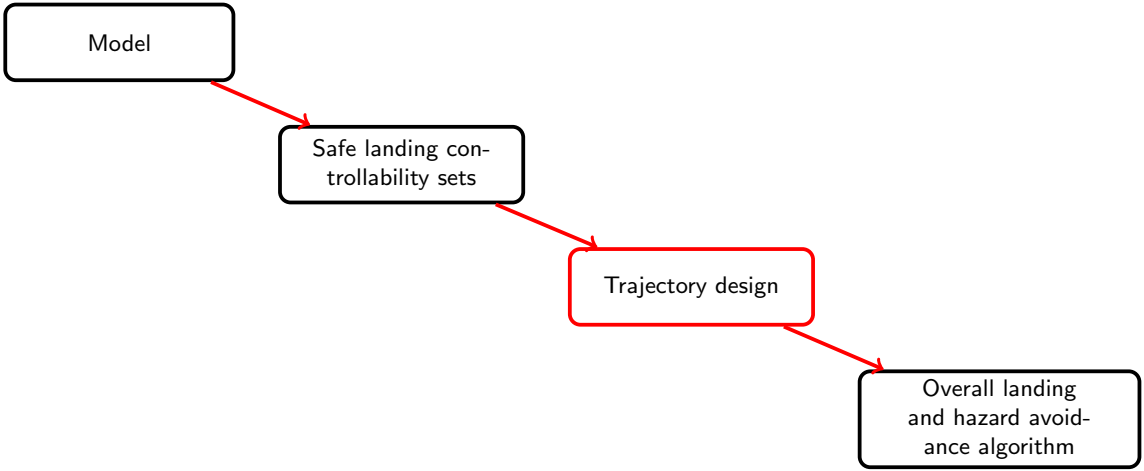
(a) $iter = 1$



(b) $iter = 2$



(c) $iter = 3$


Model
Safe landing controllability sets
Trajectory design
Overall landing and hazard avoidance algorithm

Landing trajectory design

Final model

$$S(t) = \Phi(0,t)S(0)\Phi(0,t)^\intercal, \quad S(0) = \frac{1}{R^2}\Sigma^{-1},$$
$$\dot{\bar{x}}(t) = A\bar{x}(t) + Bu(t) + G, \quad \bar{x}(0) = y(0),$$
$$\dot{m}(t) = -\beta\|T_c(t)\|$$

Constraints

- mass range
- tightened state constraints
  (maximum velocity, glideslope, landing region)
- torque bounds (nonconvex)
- pointing constraint (nonconvex)
- landing area constraint

Landing trajectory design

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$$\max_{\bar{x}(0),u,\sigma,t_f} \quad \ln m(t_f)$$
s.t. for $\forall t \in [0, t_f]$
initial condition
the convexified dynamics
the convexified constraints

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$$\text{the convexified dynamics}$$
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$\rightarrow$ **optimality and probabilistic safety guarantees**

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→ **optimality and probabilistic safety guarantees**

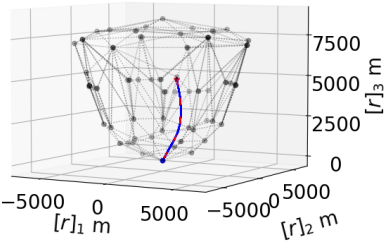
→ solves **Problem 1 (Nominal trajectory design):**
Design a tractable optimization formulation for constraint-admissible PDG under navigational uncertainty with greater than or equal to user-specified safety probability.

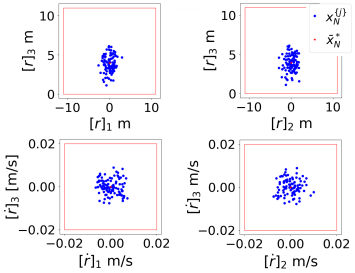
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Landing trajectory Design: Results

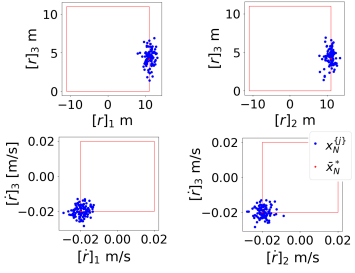
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m_{d}	900 kg	$v_{\max}$	300 m/s	α	0.7	$3\sigma([\nu_0]_{[4:6]})$	$[0.01, 0.01, 0.01]^\intercal$ m/s
ρ_1	1657.27 N	$\theta_{\max}$	25°				
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I_{sp}	255 s	γ	75°				

Landing trajectory Design: Results

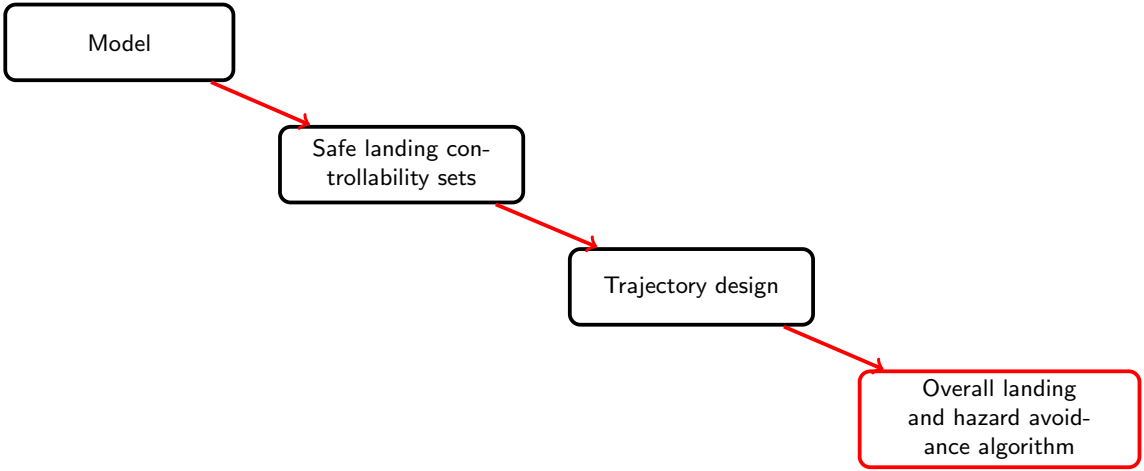




(a) With stochastic computation of the sets



(b) With deterministic computation of the sets


Model
Safe landing controllability sets
Trajectory design
Overall landing and hazard avoidance algorithm

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Final algorithm

Algorithm 0: Divert-feasible PDG under navigational uncertainty

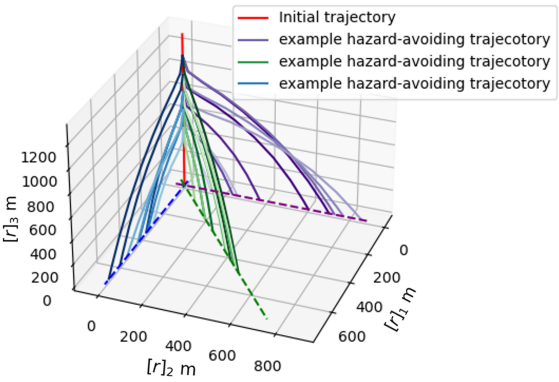
Input : x_{ls}, n_{pairs} and $(m_{\mathrm{w}}^{\mathrm{s}}, t_{\mathrm{f}}^{\mathrm{s}})$ for $s = 0, \ldots, n_{\mathrm{pairs}}$
Offline: Obtain a database of SSCS applicable to any landing location for various $(m_{\mathrm{w}}^{\mathrm{s}}, t_{\mathrm{f}}^{\mathrm{s}})$
Online:

Measure y_0 and obtain a landing trajectory and control sequence;
for $k = 0, \ldots, N-1$ **do**
    Check for hazard;
    if *hazard* **then**
        safe = False;
        $N - k \to N, \quad 0 \to k$;
        while *safe = False* **do**
            Choose a different landing site $x_{\mathrm{ls}}^{\mathrm{new}}$;
            Check database for set which contains y_0;
            If one exists set: safe=True
        Obtain new landing trajectory and control sequence with $x_{\mathrm{ls}}^{\mathrm{new}} \to x_{\mathrm{ls}}$;
Implement u_k;

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Final algorithm: Results

Moon & Lander parameters		Constraints specifications	
g	$[0, 0, -1.62]^\intercal m/s^2$	$x_{\text{ls}}^{\text{nominal}}$	$[0, 0, 0, 0, 0, 0]^\intercal$
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ρ_1	1657.27 N	$\theta_{\max}$	25°
ρ_2	4419.39 N	$\hat{n}$	$[0, 0, 1]^\intercal$
I_{sp}	255 s	γ	75°
Simulation parameters			
Δt	10 s	$3\sigma([\nu_0]_{[1:3]})$	$[3, 3, 3]^\intercal$ m
α	0.7	$3\sigma([\nu_0]_{[4:6]})$	$[0.01, 0.01, 0.01]^\intercal$ m/s

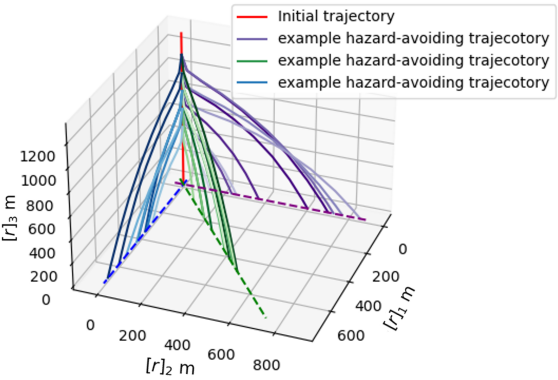


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Final algorithm: Results

Moon & Lander parameters		Constraints specifications	
g	$[0, 0, -1.62]^\intercal m/s^2$	$x_{\text{ls}}^{\text{nominal}}$	$[0, 0, 0, 0, 0, 0]^\intercal$
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Solves **Problem 2 (Hazard avoidance):**
If a hazard is detected, identify a reachable alternative landing site and use a modified version optimization-based problem to divert to the selected site



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Conclusions

Problem 1 (Nominal trajectory design):
Design a tractable optimization formulation for constraint-admissible PDG under navigational uncertainty with greater than or equal to user-specified safety probability.

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If a hazard is detected, identify a reachable alternative landing site and use a modified version optimization-based problem to divert to the selected site

→ Developed and numerically tested a **divert-feasible powered descent guidance algorithm under navigational uncertainty.**

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Thank you!

Contact: weiss@merl.com, yana.lishkova@eng.ox.ac.uk

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N. Srinivas, A. P. Vinod, S. Di Cairano, and A. Weiss, "Lunar landing with feasible divert using controllable sets," in *AIAA SCITECH 2024 Forum*, 2024, p. 0324.